

Dasari Pravallika

AI Engineer

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SUMMARY

Senior AI Engineer with around 9 years of experience in delivering production AI/ML systems for finance, health, insurance and retail. Built enterprise Agentic AI platforms that automated onboarding and ticket summarization for a major bank, cutting manual effort by 30% and handling over 10,000 daily requests. Scaled retrieval augmented generation pipelines serving millions of documents with sub second latency, and deployed them on Azure and AWS cloud platforms using Docker, Kubernetes and CI/CD pipelines. Drives cross functional collaboration to integrate Large Language Models, FastAPI services and monitoring frameworks that ensure responsible AI and operational reliability.

SKILLS

Programming & Core Libraries

JavaScript, PySpark, Python, Pandas, NumPy, SQL

Machine Learning & Deep Learning

Time Series Forecasting, Recommendation Systems, Unsupervised Learning, Hyperparameter Tuning, Supervised Learning, Anomaly Detection, Machine Learning, Model Validation, Classification, Deep Learning, Scikit learn, Regression, Clustering, TensorFlow, XGBoost, PyTorch

Generative AI & NLP

Vector Databases (Pinecone, Azure AI Search, FAISS), Retrieval Augmented Generation (RAG), Large Language Models (LLMs), Text Classification, Prompt Engineering, Transformer Models, Sentiment Analysis, Semantic Search, Azure OpenAI, Tool Calling, Agentic AI, Embeddings, LangChain, LangGraph, AI Agents, spaCy

Cloud & MLOps

Azure Kubernetes Service (AKS), Azure Synapse Analytics, Azure Machine Learning, Azure Data Factory, Containerization, Microsoft Azure, CI/CD Pipelines, Cloud Platforms, AWS SageMaker, Azure DevOps, Kubernetes, Docker, MLflow, AWS, Git, EC2, S3

Data Engineering & Databases

Exploratory Data Analysis (EDA), Statistical Analysis, Data Transformation, Feature Engineering, ETL/ELT Pipelines, Data Warehousing, Batch Processing, Data Engineering, Data Validation, Data Pipelines, Data Cleansing, Data Modeling, PostgreSQL, SQL Server, Snowflake, MySQL

APIs, BI & Development Tools

Real Time Inference, Jupyter Notebook, Batch Inference, Matplotlib, Confluence, REST APIs, Power BI, FastAPI, Tableau, Windows, Flask, Excel, Linux, Agile, Scrum, Jira

AI Governance & Monitoring

Responsible AI Practices, Operational Monitoring, Evaluation Pipelines, Telemetry Monitoring, Evaluation Scripts, Model Monitoring, Access Control, Secure SDLC, Logging

EXPERIENCE

Sr. Agentic AI Engineer | Fifth Third Bank

Jan 2024 – Present, Cincinnati, OH

- Architected and delivered four enterprise grade Agentic AI applications on Azure, leveraging Python, FastAPI, LangChain, LangGraph, and Azure OpenAI to automate end to end workflows for approximately 200 daily business processes, reducing manual effort by 35%.
- Engineered scalable Retrieval Augmented Generation pipelines with Pinecone, Azure AI Search, and 512 dimensional vector embeddings, serving over one million document queries per month and cutting knowledge base search latency by 40%.
- Created AI powered copilots and assistants in Python that onboarded 150 new hires each month, auto summarized 3,000 support tickets weekly, and provided contextual Q&A, achieving 90% first contact resolution.
- Integrated LLMs with twelve internal APIs, SQL Server data stores, and operational platforms via tool calling, automating over 30 routine tasks daily and cutting manual processing time by 28%.
- Implemented a multi agent orchestration framework using LangGraph, enabling dynamic task routing, tool calling, approval loops, and fallback handling across ten critical processes, improving workflow success rate to 96%.
- Engineered semantic search systems using embeddings, chunking strategies, metadata filtering, and vector similarity search to improve enterprise knowledge accessibility.
- Built scalable backend microservices and REST APIs using Python and FastAPI to support AI inference, workflow execution, and real time operational integrations.
- Developed and optimized AI/ML data pipelines using Python, PySpark, and SQL for ingestion, transformation, validation, and processing of large enterprise datasets.

- Applied prompt engineering techniques including reasoning chains, structured prompting, tool aware workflows, and dynamic response handling to improve AI accuracy and reliability.
- Automated operational workflows including onboarding validations, support workflows, reporting tasks, and enterprise process automation using AI driven solutions.
- Deployed all AI services in Docker containers on Azure Kubernetes Service and Amazon EKS, establishing Azure DevOps CI/CD pipelines that delivered three releases per week and achieved 99.5% uptime.
- Implemented monitoring and evaluation frameworks to track hallucination risks, groundedness, workflow performance, latency, and operational reliability across production AI systems.
- Supported production deployments, troubleshooting, backend debugging, deployment validation, workflow monitoring, and operational support activities for enterprise AI applications.
- Collaborated with product owners, platform engineers, security specialists, and business stakeholders (15+ members) in Agile sprints to define requirements, conduct code reviews, and ensure compliance, fostering cross functional delivery of AI solutions.
- Participated in Agile development processes including sprint planning, code reviews, technical discussions, testing activities, documentation, and cross functional collaboration.
- Developed AI agents capable of contextual retrieval, summarization, operational reasoning, and decision support workflows.
- Implemented comprehensive monitoring and evaluation framework using Azure Monitor and custom telemetry to track hallucination risk, latency, and workflow performance, reducing incident response time by 45%.

AI/ML Engineer (Python) | Centene Corporation

Oct 2021 – Oct 2023, St. Louis, MO

- Delivered six AI/ML solutions for claims processing, patient risk stratification, and operational reporting on Azure, processing over two million records monthly and improving claim validation accuracy by 22%.
- Built and tuned twelve predictive models with Python, Scikit learn, and TensorFlow, achieving AUC improvements of 0.07 on risk prediction and automating four high volume workflows.
- Engineered data pipelines in PySpark and SQL to ingest and cleanse three terabytes of structured and unstructured healthcare data weekly, supporting downstream ML models.
- Orchestrated end to end predictive analytics pipelines on Azure Data Factory, delivering daily risk scores for 500,000 patients and flagging 1,200 potential fraud cases per week.
- Implemented transformer based NLP pipelines with embeddings and semantic search, extracting key clinical concepts from 200,000 documents and reducing manual review time by 45%.
- Built FastAPI based AI services and REST APIs supporting healthcare analytics applications, operational reporting systems, and enterprise workflow automation.
- Integrated AI applications with healthcare databases, reporting systems, cloud platforms, and enterprise operational tools to streamline healthcare processes.
- Designed data ingestion and ETL pipelines using Python, Azure Data Factory, SQL, and PySpark for large scale healthcare data processing and transformation.
- Developed Power BI and Tableau dashboards for healthcare analytics, patient trends, claims monitoring, operational KPIs, and executive reporting.
- Applied data validation, feature engineering, preprocessing, and model evaluation techniques to improve prediction accuracy and operational reliability.
- Automated healthcare operational workflows including claims summarization, reporting automation, patient analytics, and enterprise data processing activities.
- Containerized all services with Docker and orchestrated on Kubernetes (EKS), establishing CI/CD pipelines via GitHub Actions that enabled four weekly deployments and ensured cloud platform consistency across Azure and AWS.
- Implemented monitoring frameworks for model performance, drift detection, workflow reliability, latency tracking, and operational analytics across healthcare AI systems.
- Collaborated with healthcare analysts, clinicians, product teams, data engineers, and operational stakeholders to deliver AI driven healthcare solutions.
- Supported production troubleshooting, deployment validation, backend debugging, issue resolution, and operational support for healthcare analytics and AI platforms.
- Led cross functional collaboration with clinicians, data engineers, and product managers (20+ stakeholders) in Scrum ceremonies, aligning model requirements and delivering compliant AI solutions.

Data & ML Engineer (Python) | Assurant

Oct 2019 – May 2021, New York, NY

- Designed Python ETL pipelines on Azure Data Factory processing 1.5 TB of data nightly, reducing data latency from twelve hours to two hours.
- Trained eight Scikit learn, TensorFlow, and XGBoost models for churn prediction and demand forecasting, achieving a 15% lift in forecast accuracy.

- Processed mixed structured and unstructured datasets with PySpark and SQL, delivering 250,000 feature rows per day for downstream ML pipelines.
- Designed REST APIs and backend services using FastAPI to integrate ML models with enterprise applications.
- Applied feature engineering, data preprocessing, model evaluation, and hyperparameter tuning techniques to improve model accuracy.
- Automated data validation, transformation, reconciliation, and reporting workflows to improve operational efficiency.
- Developed SQL queries, stored procedures, and backend database workflows supporting analytics and ML applications.
- Created Power BI and Tableau dashboards for operational reporting, KPI monitoring, and business analytics visualization.
- Containerized model serving APIs with Docker, deployed on Azure Kubernetes Service, and integrated CI/CD via Azure DevOps, achieving zero downtime releases for five production services.
- Facilitated collaboration across data engineering, analytics, and business teams (12 members) to define KPI dashboards, resulting in a 30% faster insight generation cycle.

Sr. Data Engineer (Python) | Finish Line

Oct 2017 – Jul 2019, Indianapolis, IN

- Designed and developed scalable data pipelines using Python, PySpark, SQL, and cloud native technologies for enterprise data processing.
- Built ETL and ELT workflows for ingestion, transformation, cleansing, and validation of structured and unstructured datasets.
- Engineered large scale batch and real time data processing pipelines using Spark, Databricks, and distributed data frameworks.
- Developed optimized SQL queries, stored procedures, data models, and backend workflows supporting enterprise analytics and reporting systems.
- Integrated data pipelines with APIs, cloud storage platforms, operational systems, and enterprise applications for centralized data access.
- Implemented data quality checks, reconciliation workflows, schema validation, and monitoring processes to ensure reliable data processing.
- Built data warehousing and Lakehouse solutions using Snowflake, Azure Synapse, Databricks, and cloud native storage platforms.
- Automated workflow orchestration and scheduling using Airflow, Azure Data Factory, and CI/CD pipelines for scalable deployments.
- Created Power BI and Tableau dashboards supporting operational analytics, KPI reporting, business intelligence, and executive reporting.
- Optimized data pipeline performance, Spark jobs, SQL processing, and cloud resource utilization for scalability and operational efficiency.
- Supported cloud migration, production deployments, troubleshooting, monitoring, and operational support activities across enterprise data platforms.
- Collaborated with data analysts, ML engineers, product teams, and business stakeholders to deliver scalable data engineering solutions.

Data Engineer (Python) | eNoah iSolution Private Limited

May 2016 – Aug 2017, Chennai, India

- Assisted in developing ETL pipelines using Python, SQL, and PySpark for data ingestion and transformation workflows.
- Processed structured and unstructured datasets from databases, APIs, and cloud storage platforms for analytics and reporting.
- Wrote SQL queries, joins, views, and stored procedures to support backend data processing and reporting activities.
- Performed data cleansing, validation, reconciliation, and transformation to ensure data quality and consistency.
- Supported batch processing and scheduled workflows for enterprise operational and analytical datasets.
- Assisted in building dashboards and reports using Power BI and Tableau for KPI monitoring and business analytics.
- Worked with cloud platforms such as Azure and AWS for data storage, processing, and workflow automation activities.
- Collaborated with data engineers, analysts, and business teams to gather requirements and support data driven solutions.
- Participated in troubleshooting, production support, deployment validation, and monitoring of data processing workflows.

EDUCATION

Bachelor of Technology (B.Tech) in Computer Science | Vasireddy Venkatadri Institute of Technology

May 2016